

AI/ML Internship – Day 47 Notes & Tasks

Module 6: Detecting Objects in Videos Using YOLO & Real-Time Webcam Detection

Part 1: Recap of Day 46

Yesterday we learned:

-  Classes
-  Bounding Boxes
-  Confidence Scores
-  Confidence Threshold
-  Reading YOLO Detection Results

Today we will learn:

 Video Object Detection

 Real-Time Webcam Detection

Part 2: What is Video Object Detection?

Video Object Detection means:

Detecting objects continuously in video frames.

Example:

Video



Frame 1 → Detect Person



Frame 2 → Detect Person



Frame 3 → Detect Car



Frame 4 → Detect Dog

YOLO processes every frame and detects objects.

Part 3: Real-World Applications

Traffic Monitoring

Detect:

- Cars
 - Buses
 - Bikes
 - Trucks
-

CCTV Surveillance

Detect:

- People
 - Intruders
 - Suspicious Activities
-

Factory Monitoring

Detect:

- Products
 - Workers
 - Defective Items
-

Smart Retail Stores

Detect:

- Customers
 - Products
 - Shopping Activities
-

Part 4: Required Libraries

Install:

`pip install ultralytics`

```
pip install opencv-python
```

Part 5: What is OpenCV?

OpenCV stands for:

Open Source Computer Vision Library

It helps us:

- ☒ Read Images
 - ☒ Read Videos
 - ☒ Access Webcam
 - ☒ Process Frames
-

Part 6: Reading a Video

Suppose we have:

video.mp4

Code:

```
import cv2
```

```
video = cv2.VideoCapture("video.mp4")
```

Part 7: Load YOLO Model

```
from ultralytics import YOLO
```

```
model = YOLO("yolov8n.pt")
```

Part 8: Detect Objects in Video

```
import cv2
```

```
from ultralytics import YOLO
```

```
model = YOLO("yolov8n.pt")
```

```
video = cv2.VideoCapture("video.mp4")
```

```
while True:
```

```
    success, frame = video.read()
```

```
    if not success:
```

```
        break
```

```
    results = model(frame)
```

```
    annotated_frame = results[0].plot()
```

```
    cv2.imshow(  
        "YOLO Detection",  
        annotated_frame  
    )
```

```
    if cv2.waitKey(1) == 27:
```

```
        break
```

```
video.release()
```

```
cv2.destroyAllWindows()
```

Part 9: How This Works

Video



Read Frame



YOLO Detection



Draw Bounding Boxes



Display Frame



Next Frame

This repeats until video ends.

📌 Part 10: Real-Time Webcam Detection

Instead of a video file, we can use the laptop camera.

Access Webcam

```
camera = cv2.VideoCapture(0)
```

Meaning

0 = Default Webcam

📌 Part 11: Webcam Detection Program

```
import cv2
from ultralytics import YOLO

model = YOLO("yolov8n.pt")

camera = cv2.VideoCapture(0)

while True:

    success, frame = camera.read()

    if not success:
        break

    results = model(frame)

    annotated_frame = results[0].plot()

    cv2.imshow(
        "Live Detection",
        annotated_frame
    )

    if cv2.waitKey(1) == 27:
        break
```

```
camera.release()
```

```
cv2.destroyAllWindows()
```

Part 12: What Happens Here?

Webcam



Capture Frame



YOLO Detection



Bounding Boxes



Display Live Output

Part 13: Example Output

Suppose camera sees:

Person

Laptop

Mobile Phone

YOLO displays:

Person 96%

Laptop 94%

Cell Phone 91%

with bounding boxes.

Part 14: Saving Detection Results

```
results = model(  
    frame,
```

```
    save=True  
)
```

Detected frames can be saved automatically.

Part 15: Performance Considerations

Faster Model

YOLO("yolov8n.pt")

Best for beginners.

Larger Models

yolov8s.pt

yolov8m.pt

yolov8l.pt

yolov8x.pt

More accurate but slower.

Part 16: Common Errors

Error 1

Camera Not Opening

Solution:

Close other applications using the camera.

Error 2

ModuleNotFoundError: cv2

Solution:

pip install opencv-python

Error 3

Video File Not Found

Solution:

Check video path.

✦ Part 17: Theory Questions

What is Video Object Detection?

Detecting objects in video frames.

What is Real-Time Detection?

Detecting objects live using a webcam.

What is OpenCV?

Open Source Computer Vision Library.

Why use YOLO for videos?

Because it is fast and suitable for real-time detection.

What does `cv2.VideoCapture(0)` do?

Accesses the default webcam.

✦ Part 18: Complete Workflow

Camera / Video



Read Frame



YOLO Model



Object Detection



Bounding Boxes



Display Output

Tasks for Day 47

Theory

1. What is Video Object Detection?
 2. What is Real-Time Detection?
 3. What is OpenCV?
 4. Why is YOLO suitable for video detection?
 5. What does VideoCapture(0) mean?
-

Practical Task 1

Install OpenCV:

```
pip install opencv-python
```

Practical Task 2

Load YOLO model.

Practical Task 3

Run detection on a video file.

Practical Task 4

Run real-time webcam detection.

Practical Task 5

Detect at least:

- Person

- Mobile Phone
- Laptop

using webcam.